# Karthikraja Natarajan

**Software Engineer · Frontend Lead · React | TypeScript | Node.js | Go**

+91 97866 10538 | pmnkarthikraja@gmail.com

## SUMMARY

Product-focused **Software Engineer with 4+ years of experience** building high-traffic web platforms, with deep expertise in **React, TypeScript and modern frontend architecture** and hands-on full-stack experience in **Node.js, Go and cloud-native deployment**. Most recently **led the frontend team for SpinPals**, a real-money gaming platform in a heavily regulated US market, owning revenue-critical systems end to end: **payments and checkout, secure onboarding, KYC and geo-compliance, and real-time features over WebSockets**. Designed the team's **CI/CD quality and security pipeline** on GitHub Actions, enabling fast and safe releases. Known for pairing strong engineering standards with a sharp focus on business outcomes: **conversion, payment success, retention and reliability**.

## SKILLS

| | |
|---|---|
| **Frontend** | React.js, Next.js, TypeScript, JavaScript (ES6+), React Native, Ionic, HTML5, CSS3, Tailwind CSS, SCSS |
| **State & Data** | TanStack React Query, Zustand, Redux Toolkit, REST APIs, WebSockets (Socket.IO) |
| **Backend & Databases** | Node.js, Express, Go, Python, PostgreSQL, MongoDB, SQLite, MySQL |
| **Cloud & DevOps** | GitHub Actions, CI/CD, Docker, AWS (Amplify, EC2, S3), Nginx, Linux |
| **DevSecOps & Quality** | gitleaks, npm audit, ESLint, Prettier, Husky, code reviews, test coverage gates |
| **Testing** | Vitest, Jest, React Testing Library, Cypress |
| **Payments & Security** | Payment gateway integration, Apple Pay, Google Pay, 3DS, OAuth 2.0, 2FA/OTP, KYC, geo-fencing |
| **Observability & Analytics** | Datadog (RUM, Logs), Mixpanel, Google Tag Manager, Meta Pixel |
| **Architecture** | Onion Architecture, Observer Pattern, PWA, code splitting, performance optimization |
| **Leadership** | Team leadership, mentoring, sprint planning, technical decision-making, Agile/Scrum, Jira |

## EXPERIENCE

**Software Engineer, Frontend Lead** - **GammaStack** 10/2025 - Present | India

*SpinPals - real-money social casino and sweepstakes platform for the US market* | **spinpals.com**

### Leadership & Engineering Excellence

- **Led the frontend team**: drove sprint planning, feature breakdown and task allocation, and acted as the go-to owner for architecture and technical decisions.
- **Mentored engineers and ran code reviews**, setting standards for clean architecture, reusable components and maintainable code.
- **Designed and implemented the CI pipeline on GitHub Actions** with parallel jobs for linting, unit tests with coverage, production build checks, complexity analysis and duplication detection.
- **Built security into the pipeline** with automated secret-leak scanning (gitleaks) and dependency vulnerability audits, catching issues before they reached production.
- **Gated pull requests only on the code each developer changed**, blocking new technical debt without stalling delivery on the legacy backlog.
- **Enforced PR governance** (branch naming conventions, required Jira ticket references, pre-commit hooks) and added a **CI metrics report** to track code-health trends run over run.

### Product & Platform Delivery

- **Owned the payments and checkout experience**, the platform's main revenue engine: integrated Coinflow, Checkout.com, Aeropay, Apple Pay and Google Pay with 3-D Secure, and built clear failure handling that **improved purchase success and reduced support load**.
- **Designed secure onboarding and authentication**: social login (Google, Apple, Facebook), OTP and phone verification, 2FA, Cloudflare Turnstile bot protection and session management.

- **Delivered compliance-first features for a regulated market**: KYC (Persona), state-level geo-fencing (Radar), fraud signals (Riskified), responsible gambling controls and redemption limits.
- **Shipped engagement and retention systems** (missions and milestones, VIP and loyalty tiers, rakeback, free spins, promotions) that drive daily active usage and player lifetime value.
- **Built real-time features over WebSockets**: live jackpots, wallet balance sync and in-app notifications.
- **Improved production reliability** with Datadog RUM, logging, source-map pipelines and error boundaries, cutting time to find and fix production issues.
- **Built growth and analytics tooling** (Mixpanel, Meta Pixel advanced matching, GTM) for accurate marketing attribution and data-driven product decisions.
- Built a **performant, mobile-first PWA** with code splitting, lazy loading, service workers and PixiJS canvas animations.

**Stack:** React 18, Vite, TanStack Query, Zustand, Socket.IO, Tailwind CSS, Vitest, GitHub Actions, Docker, Nginx, Datadog

**Software Engineer** - **Datasack Solutions** 05/2024 - 09/2025 | Tamil Nadu, India

- **Independently designed and built a full-stack Payroll and HR platform** for Saudi-based organisations, with three connected portals for employees, admins and payroll.
- Automated the request-to-payroll flow (leave, allowances, advances, claims) with role-based approvals and PDF payslip generation, **reducing payroll processing errors by over 75%** for 300-400 employees.
- Deployed on **AWS Amplify, EC2 and S3 with Docker** and CI/CD pipelines, using Onion Architecture and the Observer Pattern for a maintainable, testable codebase.
- Built an **SEO-optimised marketing site with a chatbot and lead capture**, generating 5-10 qualified inquiries per month through organic traffic.
- Built a cross-platform **CRM reminder app for Android and iOS** from a single codebase with React Native and Ionic.
- Wrote unit, integration and end-to-end tests with Jest and Cypress, and took part in regular code reviews and refactoring.

**Stack:** React.js, Next.js, TypeScript, Node.js, MongoDB, Tailwind CSS, Elastic UI, React Native, Docker, AWS Amplify

**Junior Software Engineer** - **Midaas Telesoft Pvt. Ltd.** 08/2022 - 03/2024 | Gujarat, India

- **Built real-time industrial monitoring dashboards** using the Observer Pattern over WebSockets, with TanStack Query for efficient server-state management.
- **Developed a rendering engine for dynamic Single Line Diagrams** using ReactFlow and PixiJS, visualising live changes across complex electrical systems.
- Implemented and extended **industrial IoT protocols (DNP3, Modbus, IEC 60870)** in Go over serial and TCP, including decoding binary sensor data for reliable readings.
- Automated testing with Cypress alongside unit and integration testing.

**Stack:** Go, TypeScript, React.js, Node.js, WebSockets, ReactFlow, PixiJS, Docker, Cypress, Linux

**Project Trainee** - **Indian Institute of Science (IISc)** 12/2021 - 06/2022 | Bengaluru, India

- Built a web-based computing server that finds Pi-Pi interactions in protein structures from the Protein Data Bank, with data visualisation for researchers, deployed publicly on the web.

**Stack:** Python (CGI), JavaScript, HTML5, CSS, MySQL, Apache, Linux

## EDUCATION

**Master of Science, Bioinformatics** - Bharathidasan University | CGPA 8.6 2017 - 2022

## CERTIFICATION

**Micro1 Certified Developer** - passed a technical assessment in full-stack development covering coding, problem-solving and an AI interview.